

A Map of Tax Responsiveness

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Preliminary and incomplete. The latest version is available [here](#).

Abstract

The questions economists ask of tax elasticities are seldom local, yet the methods used to estimate them rest on local, *between-income* contrasts: taxpayers at a kink, or a reform-affected group against a nearby unaffected one. The variation examined in this paper comes, instead, from differential, *within-income* shifts to the tax liability function. This makes it possible to control for income without absorbing identifying variation, strengthens monotonicity and exogeneity – but also enables, otherwise unfeasible, heterogeneity analyses across the income distribution, spousal income split and gender. Applying the design to the universe of French income tax returns from 2003 to 2023, we find elasticities of earnings that decline with earnings – most steeply at low incomes – a gradient driven by low earners insured by higher-earning partners rather than by those in households where both partners earn little. The answers to normative questions – whom to tax more, whom less – and to positive ones – what are the effects of tax reforms, e.g. towards individual taxation – are sensitive to this heterogeneity.

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1 Introduction

When a picture contains many moving parts, it can help to zoom in. To measure tax responsiveness in tax data, the main problem is simultaneity: tax rates are a function of income, and income, a function of how much it gets taxed, which means that a naive regression of income growth on tax-rate changes does not give a causal estimate of a “tax elasticity”. In response, the literature has gone local: (i) zooming in on a kink in the tax schedule, inferring tax elasticities from the excess density of taxpayers just below a threshold—the idea behind bunching designs (H. J. Kleven, 2016; Saez, 2010); or (ii) zooming in on a group affected by a tax reform, approximating its counterfactual earnings growth with a similar group left unaffected—the idea behind simulated tax-reform IV designs (Gruber & Saez, 2002).

The questions for which elasticity-estimates are needed, however, often take the zoom out. The tax elasticity of labour supply (or, “sufficiently”, of taxable income; Feldstein, 1999) is a key input to optimal-tax models, macroeconomic labour supply models, and microsimulations of the effects of reforms – and it may matter which local estimate is drawn: an elasticity identified from a cut to top rates need not be close to any representative, relevant average. And many questions are comparative: the costs of a progressive tax schedule depend on the gradient of responsiveness across the income distribution, and the costs of a joint tax schedule (taxing married couples’ income as one), on the ratio of spousal elasticities. The variance of existing estimates is large (Neisser, 2021), and it is difficult to establish how much of it reflects actual heterogeneity in tax responsiveness, and how much the populations, margins, and designs that differ across studies.

This paper proposes a source of variation suited to identifying heterogeneities across income, spousal income split and gender. Like bunching, the approach exploits discontinuities in the tax schedule, but it does so in the context of “events” that change the tax liability function – such as, under family taxation, the fiscal exit of a child from the family tax unit or marriage. It relies on the fact that, under the French system, differences in partner income make otherwise identical individuals differentially likely to switch tax bracket: two identical women with the same earnings, e.g. both in the 14% bracket, whose first-born reaches the age at which fiscal unit attachment ends, may see different tax changes – one remaining at 14%, the other, with an

a bit higher-earning partner, pushed into the 30% bracket. The idea is to use the assumption that there is no anticipatory bunching around not yet effective (childless or joint) thresholds.

Two features of this variation make it interesting. (i) It comes from outward shifts of otherwise unchanged tax schedules, which makes the monotonicity assumption more plausible than under reform variation. (ii) It is within- rather than between-income. When exposure to tax variation is a function of own income – as with a reform affecting a particular part of the income distribution – income cannot be controlled for without absorbing identifying variation, leaving between-income designs susceptible to bias from mean reversion and secular trends in income. These are arguably less of a concern when the source of variation is not income but partner income differences. And because it is present throughout the own- and partner income distribution, some otherwise difficult heterogeneity analyses become feasible – in one observational setting.

For such multi-dimensional heterogeneity analysis, correspondingly large data is needed. From the universe of French tax returns from 2003 to 2023 – filings at the tax-unit level – we built an individual-level panel.¹ It is the size² of the data that permits a very granular analysis: *within* each narrowly defined cell of the own and partner income distribution, we analyse every possible event-induced tax change separately and then aggregate – and do so by gender and event type.

We find elasticities that decline with income – most steeply at low incomes – and are shifted upward for women; the gradient is driven by low earners insured by higher-earning partners rather than by those in households where both partners earn little.

We show the estimated heterogeneity to alter the conclusions of normative models of optimal taxation and of reform assessments. Feeding the estimated income profile into the sufficient-statistics formula for optimal taxation (Saez, 2001), in the implementation of Brewer et al. (2010), makes the optimal schedule substantially steeper than its constant-elasticity benchmark; replacing the constants in the evaluation of reforms towards more individual taxation of couples (Bierbrauer et al., 2024) raises the assessed welfare gains (Section 6).

¹Following individuals through events such as marriage requires linking them across tax units, which individual-level identifiers in the data permit.

²Around 50 million observations per year.

The paper thereby joins a long-standing effort to understand the dimensions of heterogeneity in tax responsiveness. One dimension is the margin of adjustment — taxable income, broader income concepts, labour supply — where the literature can be summarised as finding that the more comprehensive the income definition considered, the larger the elasticity, with work-income-only elasticities for men barely distinguishable from zero. Another dimension is the income distribution: studies of multiple reforms (Gruber & Saez, 2002; Jakobsen & Søgaaard, 2022; H. J. Kleven & Schultz, 2014) or of reforms spanning wide income ranges (Graber et al., 2026) generally find larger broad-income responses at the top than in the middle of the distribution. Papers on heterogeneity across gender find married women and single mothers to be above-average elastic in labour supply (Blomquist & Selin, 2010; Eissa, 1995, 1996), and a further strand studies heterogeneity over time, arguing that the gap between macro and micro estimates hinges on different horizons considered (H. Kleven et al., 2025).

The paper also contributes to the methodological effort to clarify the conditions under which, and precisely what, simulated-instrument strategies identify. Gruber and Saez (2002), Weber (2014), Blomquist and Selin (2010), H. J. Kleven and Schultz (2014), and Miao et al. (2024) propose ways to mitigate biases from mean reversion and inequality trends, whose respective merits are discussed by Jakobsen and Søgaaard (2022) and Saez et al. (2012); Kumar and Liang (2020) and Graber et al. (2026) relax the constant-elasticity assumption, and Gruber and Saez (2002) and Graber et al. (2026) separate income from substitution effects. We draw most on Jakobsen and Søgaaard (2022), whose strategies for validating the identifying assumptions we adapt.

The rest of the paper is structured as follows. Section 2 describes the French tax system and the construction of the individual-level panel. Section 3 sets out the theoretical framework, the identification problem, and the event-based instruments we propose. Section 4 develops the empirical strategy and illustrates it with a transparent case study, Section 5 presents the results, and Section 7 concludes.

2 Background

2.1 Institutional Setting

The variation explored in this paper stems from two features of the French tax system: (i) the dependence of the tax liability on household composition through the *family quotient*, and (ii) the step-wise shape of the marginal tax schedule. Together, they imply that a change in household composition can move a household across a bracket threshold—and whether it does depends on the incomes of both spouses.

What makes the French tax system one of “family taxation” is that the tax burden (equation (1)) is a function of the family quotient, which increases with the number of family members: the quotient is 1 for singles, 2 for childless married (or civil-union) couples and 3 for a married couple with two children, increasing for each child up to a cap.³⁴

$$(1) \quad \text{Tax burden} = T\left(\frac{\text{Taxable income}}{\text{Family quotient}}\right) \times \text{Family quotient}$$

where $T(x)$ denotes the French income tax formula for a single individual.

The resulting marginal tax rates for different family compositions are depicted in Figure 2 Panel A. Panel B illustrates the resulting income tax liabilities, accounting for both the family quotient and the cap on child-related tax advantages.

³The tax advantage associated with children is capped, such that the benefit from an additional child does not exceed a predetermined threshold.

⁴Other factors also influence the family quotient. For example, it increases for single parents and parents of disabled children. We do not consider these cases in our analysis.

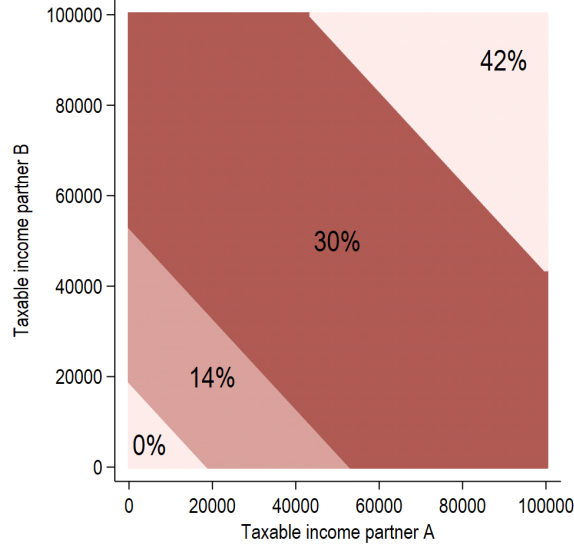
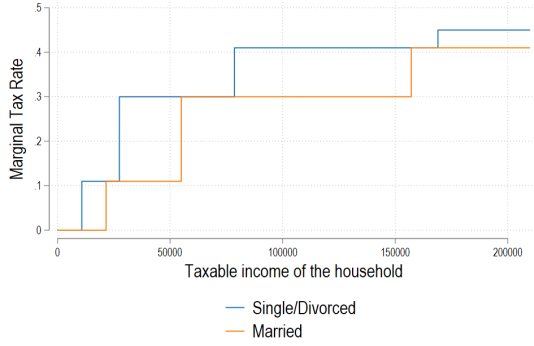
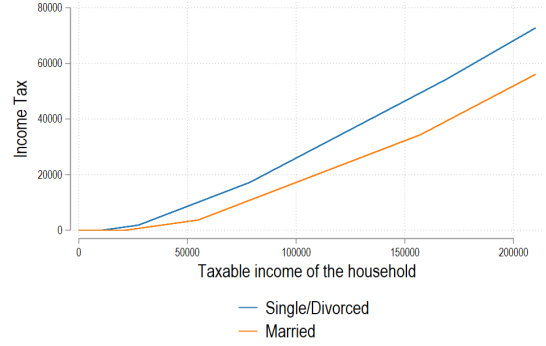


Figure 1: Marginal tax rate as a function of the income of both spouses

Notes: The heatmap illustrates how the marginal tax rate depends on the income of both partners. Conditional on the income of one partner, the marginal tax rate increases with the income of the other partner. The simulation uses the French tax system of 2015.



(a) Marginal tax schedule



(b) Tax burden

Figure 2: Marginal tax schedule and tax burden by household composition

Notes: Panel (a) displays marginal tax rates for different household compositions in 2015. Panel (b) displays the tax burden for different family situations in 2015. Source: Own calculations based on OpenFisca France.

Under French family taxation, the tax function is defined at the household level. In the notation of equation (1), a married or civil-union household with family quotient Q_{it} owes

$$T_t^H(z_{it} + z_{pt}; Q_{it}) = Q_{it} T_t\left(\frac{z_{it} + z_{pt}}{Q_{it}}\right),$$

where z_{pt} is the partner’s income. Two features drive everything that follows. First, the marginal rate is common to both spouses:

$$\frac{\partial T_t^H}{\partial z_{it}} = \frac{\partial T_t^H}{\partial z_{pt}} = T_t' \left(\frac{z_{it} + z_{pt}}{Q_{it}} \right), \quad \tau_{it}^H \equiv 1 - T_t' \left(\frac{z_{it} + z_{pt}}{Q_{it}} \right),$$

so that i ’s marginal net-of-tax rate depends on partner income—as illustrated in Figure 1—and is an equilibrium object, jointly determined with both spouses’ income choices. Second, events that change the household’s fiscal configuration shift this function at every level of own income: the fiscal exit of a child lowers Q (from 2.5 to 2 for a one-child couple, Figure 2), and marriage pools the tax base. These are the shocks our design exploits.

2.2 Data

We use the French administrative tax panel known as POTE, provided through secure access by the CASD (Centre d’accès sécurisé aux données).⁵ The dataset covers all French income tax returns between 2003 and 2023.

POTE includes all variables submitted in the tax return (Form No. 2042), intermediate calculations for determining the final tax liability, and all information shown to the household on its official tax notice.⁶

All residents in France are required to file an annual tax return reporting income and family-related information. This results in a total sample of approximately 50 million individuals per year. Married couples and individuals in civil unions (PACS) must file jointly, except in the year of marriage, when they may opt to file separately.

The dataset contains rich tax-relevant information, including—at the individual level—age, gender, work income, deductible expenses and calculated tax liabilities. It also includes detailed household-level characteristics such as the number of dependent children, marital status and filing status.

Although the returns are filed at the tax unit level (the *foyer fiscal*), the POTE database has the important advantage of including individual-level identifiers, which - to our knowledge -

⁵POTE – Panel On The Taxation of Households, INSEE–DGFIP. Data accessed via CASD (<https://www.casd.eu/>).

⁶From 2016 onward, this includes the marginal tax rate explicitly communicated to the household.

we are the first to take advantage of. It permits the construction of a longitudinal individual-level panel, essential to our design: it lets us link individuals across tax units and follow their incomes before and after family events that change the tax burden, such as marriage.

3 Estimating tax elasticities

In this section, we (i) set out the theoretical framework and define the estimand, (ii) introduce simulated tax instruments—the reform-based version used in the literature and the event-based version we propose—within that framework, and (iii) extend the framework to a two-earner household to make precise what our variation identifies for individuals in couples. Section 4 then develops the empirical strategy and illustrates it with a transparent case study.

3.1 Standard theoretical framework

We follow Jakobsen and Sogaard (2022) and specify a quasi-linear utility function for individual i in year t :

$$u_{it}(c_{it}, z_{it}) = c_{it} - \frac{n_{it}}{1 + \frac{1}{\epsilon}} \left(\frac{z_{it}}{n_{it}} \right)^{1 + \frac{1}{\epsilon}},$$

where c_{it} is consumption and z_{it} is (work / taxable) income; the second term captures the disutility of earning income, governed by the elasticity parameter ϵ and potential income n_{it} .

The budget constraint is

$$c_{it} \leq z_{it} - T_t(z_{it}; Q_{it}),$$

where the tax schedule T_t may depend on household composition through the family quotient Q_{it} of equation (1); for a single filer, $Q_{it} = 1$ and T_t is the individual schedule. Potential income may vary with family events: we write $n_{it} = n_{it}(g_{it})$, where $g_{it} \in \{0, 1\}$ indicates whether the event under study—in our design, the fiscal exit of the first-born child; marriage in the robustness design—has occurred by year t , and leave this dependence unrestricted.⁷

Changes in consumption needs associated with the event enter the budget constraint as a

⁷The occurrence and timing of the event are not modelled as choices: the event year m_i —the calendar year in which the first-born turns 21—is fixed by fertility two decades prior. The remaining institutional choice margins (continued fiscal attachment of students, the timing of detachment) are discussed alongside the density tests of Appendix B. For tax reforms, no dependence of n_{it} on the policy change needs to be specified.

lump-sum term and, with utility linear in consumption, leave the earnings choice unaffected. We record this property as an assumption, since it is what the event-based design must defend:

Assumption S1 (No income effects). *Utility is quasi-linear in consumption: earnings respond to the marginal net-of-tax rate only, and lump-sum changes in tax liability or consumption needs leave the earnings choice unaffected.*

Appendix A relaxes Assumption S1; needs and tax liabilities then enter the first-order condition through the marginal utility of consumption. Section 4 restates Assumption S1 in potential-outcomes form (Assumption 2). The first-order condition is

$$z_{it} = n_{it}(g_{it}) (\tau_{it})^\epsilon, \quad \tau_{it} \equiv 1 - T'_t(z_{it}; Q_{it}),$$

where τ_{it} is the marginal net-of-tax rate. Taking logs and first differences yields the estimable equation

$$(2) \quad \Delta \ln z_{it} = \epsilon \Delta \ln \tau_{it} + \Delta \ln n_{it}(g_{it}),$$

where $\Delta \ln n_{it}(g_{it})$ captures changes in potential income, including any direct effect of the event. The parameter ϵ is the estimand: the elasticity of (taxable or work) income with respect to the marginal net-of-tax rate.⁸ Section 3.3 makes precise what it measures for individuals in couples.

3.2 Standard variation and our variation

Since the marginal tax rate is a function of income, $\Delta \ln \tau_{it}$ in equation (2) is endogenous: any shock to income moves the individual along the progressive schedule. The literature addresses this endogeneity with *simulated instruments*: mechanical predictions of the tax change, computed by applying a change in the tax environment to predetermined income (Gruber & Saez, 2002); see Saez et al. (2012) for an overview. Designs differ in what changes the tax environment.

⁸We suppress an individual subscript on ϵ in this section for readability. Section 4 treats the elasticity as individual-specific (ζ_i) and characterises which weighted averages of the individual parameters our estimands recover.

Reform-based instruments. In the standard design, the change in the tax environment is a reform, and the instrument applies the post- and pre-reform schedules to pre-reform income,

$$\Delta \ln \tau_{it}^{\text{reform}} = \ln(1 - T_t^{\text{post}}(z_{i,t-k})) - \ln(1 - T_t^{\text{pre}}(z_{i,t-k})).$$

Validity requires the simulated change to be unrelated to potential-income growth conditional on controls,

$$(3) \quad \text{cov}(\Delta \ln \tau_{it}^{\text{reform}}, \Delta \ln n_{it} \mid z_{i,t-k}) = 0,$$

read as parallel income paths absent the reform. The assumption is demanding because exposure to reforms is usually *between*-income (Jakobsen & Sogaard, 2022): the instrument is a function of lagged income, and potential-income growth is related to the level of income through mean reversion and differential secular income trends, the two leading confounds of this literature.⁹ A further concern, developed in Appendix F, is that reform variation can move different individuals’ realised marginal rates in opposite directions—income effects and extensive-margin responses being the leading channels—in which case the IV estimand no longer aggregates behavioural responses with convex weights.

Event-based instruments. We propose to let the change in the tax environment be a family event that changes the household’s *fiscal configuration*: the set of persons whose incomes are pooled in the tax base, and the family quotient dividing it (Section 2.1).¹⁰ Writing $\mathcal{U}_d$ for the tax unit and Q_d for the quotient before ($d = 0$) and after ($d = 1$) the event—so that d

⁹See Jakobsen and Sogaard (2022) for a detailed discussion of equation (3) and of these two channels.

¹⁰Children are part of the parents’ tax unit through age 17 automatically; young adults can remain fiscally attached up to age 21 (25 if in education). Upon detachment, parents may instead deduct support payments (*pension alimentaire*) up to a cap. These choice margins mean the timing of the fiscal exit is not fully mechanical; Appendix B probes manipulation directly with density tests, and pre-trends provide a further check.

coincides with the event indicator g of Section 3.1—per-part taxable income in year t is

$b_{it}(d) = \frac{\sum_{j \in \mathcal{U}_d} z_{jt}}{Q_d},$

Event	$d = 0$		$d = 1$	
	$\mathcal{U}_0$	Q_0	$\mathcal{U}_1$	Q_1
Child’s fiscal exit: one dependent child	$\{i, p\}$	2.5	$\{i, p\}$	2
two dependent children	$\{i, p\}$	3	$\{i, p\}$	2.5
Marriage (childless couple)	$\{i\}$	1	$\{i, p\}$	2

The household’s pre-event constellation—the number of dependent children, single or joint filing—selects the relevant row and is what Section 4 records as the event type e_i . The instrument is the change in the log net-of-tax rate predicted by the event alone, obtained by applying the pre-event year’s schedule to pre-event incomes under the two configurations:

$$(4) \quad Z_i \equiv \ln(1 - T'_{m_i-1}(b_{i,m_i-1}(1))) - \ln(1 - T'_{m_i-1}(b_{i,m_i-1}(0))),$$

where m_i is the calendar year of the event; for the fiscal exit, m_i is the year in which the first-born turns 21, fixed by fertility two decades prior (Section 3.1). By the common-marginal-rate property of Section 2.1, Z_i is expressed in the units of the household net-of-tax rate τ^H , the first-stage outcome of Section 4. Whether the event is predicted to change a household’s marginal rate, and by how much, depends on where predetermined income sits relative to the thresholds of the post-event schedule—for given own income, on partner income (Figure 1). This is *within*-income variation:¹¹ among individuals with the same own income, the same pre-event bracket, and an event of the same type in the same year, some households are pushed over a bracket threshold and others are not, depending on the partner’s income. At the child’s fiscal exit, thresholds contract and the marginal rate weakly rises at every level of household income, so $Z_i \leq 0$: the event predicts tax increases only. At marriage, the tax base pools and predicted increases and decreases coexist. Table 1 illustrates.

¹¹Tax variation among taxpayers at the same income level is showcased by H. J. Kleven and Schultz (2014), whose Danish reforms create “variation even for taxpayers at the same income level”; Jakobsen and Sogaard (2022) develop the associated identifying assumption of constant trend differentials and its graphical validation.

Table 1: Two stylised couples in the year before the first-born turns 21

	Woman A	Woman B
Own income	20,000	20,000
Partner income	35,000	40,000
Bracket at $Q = 2.5$	14%	14%
Predicted bracket at $Q = 2$	14%	30%
Instrument Z_i (equation (4))	0	-0.206

Notes: The table shows two households identical in the woman’s own income and pre-event tax bracket, whose predicted tax treatment upon the first-born’s fiscal exit differs because of partner income. Brackets refer to the 2015 schedule; $-0.206 = \ln(0.70/0.86)$.

The validity requirement for the event-based instrument parallels equation (3): the predicted change Z_i must be unrelated to potential-income growth among comparable households,

$$(5) \quad \text{cov}(Z_i, \Delta \ln n_{it} \mid z_{i,m_i-1}, m_i, e_i) = 0.$$

The comparison with the reform design clarifies why the event variation is better placed to meet it. Exposure to reforms is between-income: the instrument varies with the level of lagged own income, precisely the dimension along which mean reversion and secular income trends operate. Exposure to the events is within-income: conditional on own income, the event year, and the event type, Z_i varies only with partner income, so the confounds tied to the level of own income are held fixed by design, and the identifying restriction concerns partner income within narrow bands. Events, unlike reforms, have direct effects on potential income— n_{it} depends on g_{it} —but predicted controls experience the same event, so these direct effects difference out within cells. Section 4 states the within-band restriction as Assumption 1—conditional common trends within cells—and characterises which averages of the individual elasticity parameters the resulting estimands recover.

3.3 Extended theoretical framework: the household

Equation (2) is written for one person. Pairing it with marginal rates produced by a household tax system is standard practice,¹² but under family taxation it leaves the interpretation of ϵ imprecise: partner income enters τ_{it} , and the partner responds to the same rate. We there-

¹²Most prominently, H. J. Kleven and Schultz (2014) estimate the one-person model on Danish administrative data in which, despite individual filing, “for a married person, income tax liability depends on both individual incomes and on spousal incomes” (their footnote 6), owing to exemptions transferable across spouses; their

fore extend the model to a household choosing both spouses' incomes against a joint budget constraint,

$$\max_{z_{it}, z_{pt}} c_t - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}; Q_{it}),$$

where z_{pt} is the partner's income, $T_t^H(z_{it} + z_{pt}; Q_{it}) = Q_{it} T_t((z_{it} + z_{pt})/Q_{it})$ is the household tax function of equation (1), and Ψ is a convex joint cost of earning income. The joint cost may be additively separable across spouses—the specification underlying most quantitative work on couples taxation (Bick & Fuchs-Schündeln, 2018)—or non-separable, reflecting leisure complementarities, home production, or coordinated retirement (Gelber, 2014; Goux et al., 2014).¹³

Because the household tax base pools incomes, the marginal rate is common to both spouses (Section 2.1), $\partial T_t^H / \partial z_{it} = \partial T_t^H / \partial z_{pt}$, and we write $\tau_{it}^H \equiv 1 - T_t'((z_{it} + z_{pt})/Q_{it})$ for the common net-of-tax rate, an equilibrium object jointly determined with both spouses' choices. The first-order conditions, $\partial \Psi / \partial z_{it} = \partial \Psi / \partial z_{pt} = \tau_{it}^H$, log-linearise to

$$(6) \quad \Delta \ln z_{it} = \epsilon_{ii} \Delta \ln \tau_{it} + \epsilon_{ip} \Delta \ln \tau_{pt} + \Delta \ln n_{it},$$

where τ_{it} and τ_{pt} are the spouse-specific marginal net-of-tax rates and the first subscript of each elasticity indexes whose earnings respond, the second whose rate changes: ϵ_{ii} is the own-rate elasticity of i 's earnings, and ϵ_{ip} is the cross-rate elasticity—the response of i 's earnings to a change in the *partner's* rate, transmitted, under S1, entirely through the partner's behavioural adjustment. The analogous equation for $\Delta \ln z_{pt}$ carries ϵ_{pp} and ϵ_{pi} . A non-zero cross-rate elasticity requires non-separable effort costs, $\partial^2 \Psi / \partial z_{it} \partial z_{pt} \neq 0$.

What matters for interpretation is whether the partner's rate changes along with one's own. Under joint filing the two rates coincide period by period, so any event-induced change is

tax simulator accounts for this jointness. The couples dimension is incidental to their reform-based variation, whereas it is our identifying variation, which is why we make the household structure explicit.

¹³Quasi-linearity in household consumption carries Assumption S1 to the household: it shuts down income effects, so that neither the partner's net income nor the change in the household's tax *liability* affects earnings choices, and it makes the intra-household allocation of consumption irrelevant for earnings: with transferable utility, any sharing rule supported by lump-sum transfers leaves the first-order conditions unchanged. For the same reason the model's predictions for earnings coincide with those of an efficient collective household à la Chiappori with quasi-linear private utilities—we need not take a stance on the household decision process. See Appendix A for the extension with income effects.

common to both spouses, $\Delta \ln \tau_{it} = \Delta \ln \tau_{pt} = \Delta \ln \tau_{it}^H$, and (6) collapses to

$$\Delta \ln z_{it} = \underbrace{(\epsilon_{ii} + \epsilon_{ip})}_{\equiv \epsilon_i^H} \Delta \ln \tau_{it}^H + \Delta \ln n_{it}.$$

For couples, the coefficient ϵ in equation (2) is therefore to be read as $\epsilon_i^H = \epsilon_{ii} + \epsilon_{ip}$ whenever both spouses' rates move together—as they always do at the child's fiscal exit: the *total* elasticity of own earnings with respect to the household net-of-tax rate, including the component transmitted through the partner's response. At marriage, by contrast, the two spouses' rate *changes* generally differ—the pre-marriage rates were individual—and in some cells the partner's rate does not change at all; we exploit this below. The income-weighted sum of ϵ_i^H across spouses is the household taxable-income elasticity relevant for revenue and Laffer calculations, however the response splits across own- and cross-rate channels. We estimate equation (2) separately by gender, allowing ϵ_i^H to differ across spouses.

Existing evidence suggests that cross-rate elasticities are small: exploiting the Swedish move towards individual taxation, in which spouses' marginal rates moved independently, Gelber (2014) estimates compensated cross-rate elasticities of approximately 0.05, an order of magnitude below typical own-rate estimates. The separable specification—equivalently, the one-person model applied spouse by spouse—is therefore a good approximation. We note that the distinction is not specific to our design: whenever a reform changes both spouses' marginal rates, reform-based estimates likewise recover combinations of own- and cross-rate responses, a point the literature has left implicit. Moreover, the approximation can be dispensed with: ϵ_{ii} can be isolated (i) from marriage cells in which the partner's rate is unchanged ($\Delta \ln \tau_{pt} = 0$), where the reduced form identifies ϵ_{ii} alone, so that the comparison with common-change cells identifies ϵ_{ip} ; or (ii) by imputing $\hat{\epsilon}_{ip}$ from external estimates.¹⁴

¹⁴In more detail, there are two routes from the identified total to the structural own-rate elasticity $\epsilon_{ii} = \epsilon_i^H - \epsilon_{ip}$. **Route 1** (*spousal separability, S2*): assume $\partial^2 \Psi / \partial z_{it} \partial z_{pt} = 0$, so that $\Psi = \psi(z_{it}; n_{it}) + \psi(z_{pt}; n_{pt})$. Then $\epsilon_{ip} = 0$ and $\epsilon_i^H = \epsilon_{ii}$; specialising ψ to the iso-elastic form, the household problem separates into two copies of the one-person model, each facing the household rate—the formal justification for the standard practice of pairing the one-person model with household-rate variation. **Route 2** (*external adjustment*): keep $\epsilon_{ip} \neq 0$ and impute it from Gelber (2014); the structural elasticity is recovered as $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$, reported as an adjustment or as bounds. Three remarks discipline the latter exercise. The sign: leisure complementarity implies $\epsilon_{ip} > 0$, so the unadjusted ϵ_i^H *overstates* the structural elasticity. The magnitude: plausible values of ϵ_{ip} are an order of magnitude below our estimates, so the adjustment is second-order for our conclusions. The scope: external estimates are compensated, gender-specific, and from a different population and margin, so we treat them as orders of magnitude rather than point corrections, and apply the adjustment to the intensive margin only—for participation, external cross-elasticity estimates are scarce and the added-worker

Two extensions of the model, developed in Appendix A, are used below. Allowing for income effects—a concave utility of consumption—adds a virtual-income term to equation (2); absent income effects, compensated and uncompensated elasticities coincide, while in their presence the estimands recover a total response to the event-induced change in the schedule rather than either structural elasticity, and the design’s distance-to-threshold variation separately identifies the income effect (Section 4, Assumption 2; Appendix A). And a participation-margin block in the spirit of H. J. Kleven et al. (2009) rationalises the extensive-margin analysis: there, the relevant price is the participation tax wedge, whose first-euro marginal rate is exactly what partner income and the fiscal configuration determine under the quotient system.

4 Empirical strategy

Section 4.1 introduces the comparisons that underlie the design and illustrates them with a case study. Section 4.2 places these comparisons in a potential-outcomes model and states the identifying assumptions; Section 4.3 shows that under these assumptions the resulting estimands are positively weighted averages of individual elasticity parameters. Sections 4.4 and 4.5 describe how the cell-level estimates are aggregated into the paper’s target parameters, and how estimation and inference proceed.

Notation For each individual i experiencing the event in calendar year m_i , we construct an event window around m_i , with event time $r \equiv t - m_i$, a pre-period ($r = -3, -2, -1$) and a post-period ($r = 0, \dots, R$), with $R = 3$. Outcomes are measured relative to the pre-event year: earnings growth $\Delta y_{ir} \equiv \ln z_{ir} - \ln z_{i,-1}$ and net-of-tax rate growth $\Delta \text{NTR}_{ir} \equiv \ln \tau_{ir}^H - \ln \tau_{i,-1}^H$, with τ^H the household net-of-tax rate of Section 3.3.

Why not one comparison A possible estimand for the tax elasticity of earnings is the difference-in-differences estimand that, at a given horizon r (suppressed), compares the earnings and net-of-tax rate growth of treated and control households, with Z_i the event-predicted

channel can reverse the sign. The distinction matters most for the heterogeneity results: gender differences in ϵ_i^H are preference parameters only under Route 1, or after the Route 2 adjustment; without either, a larger measured response of one spouse may partly reflect the other spouse’s adjustment transmitted within the household. Because we estimate both spouses’ responses to the same household shock within the same cells, the pair of estimates is itself informative about such reallocation, and we return to this when interpreting the heterogeneity results.

rate change of equation (4):

$$\beta^{DD} \equiv \frac{\overbrace{\mathbb{E}[\Delta y \mid Z \neq 0] - \mathbb{E}[\Delta y \mid Z = 0]}^{\equiv RF \text{ (Earnings DiD)}}}{\underbrace{\mathbb{E}[\Delta NTR \mid Z \neq 0] - \mathbb{E}[\Delta NTR \mid Z = 0]}_{\equiv FS \text{ (Net-of-tax rate DiD)}}}.$$

Two concerns arise with this estimand. (i), Z takes many values: pooling compares households experiencing tax changes of different sizes, which can place negative weights on individual responses when elasticities are heterogeneous (Callaway et al., 2024; Graber et al., 2026). (ii), the estimand requires treated and control households to share the same counterfactual earnings growth. Unconditionally, the two groups differ in own income and in partner income, the dimensions along which mean reversion and secular trends may operate.

4.1 Our approach, illustrated with a case study

We address concern (i) by comparing each dose δ in the support of Z separately to the zero-dose group, and concern (ii) by making these comparisons within cells of individuals alike in the determinants of income growth: own income within narrow bins, the event year and the event type fixed exactly, and partner income restricted to an interval. Formally, let $X_i \equiv (z_{i,-1}, z_{p,-1}, m_i, e_i)$ collect these predetermined variables, let the cells $\mathcal{I}_c$ be the elements of the resulting partition of the support of X , and write $c(X_i)$ for the cell of individual i .¹⁵ Throughout, expectations subscripted by c , $\mathbb{E}_c[\cdot]$, are taken over the individuals of cell $\mathcal{I}_c$.

Consider one such cell: individuals with pre-event own income between €34,000 and €36,000, a pre-event marginal rate of 14%, and 2.5 fiscal parts—a married or civil-union couple whose one dependent child exits the tax unit at $r = 0$. Within this cell, individuals whose partner earns little enough that per-part income stays below the threshold once the quotient falls to 2 have $Z_i = 0$; individuals with higher-earning partners are pushed into the 30% bracket and have $Z_i = \delta < 0$. Table 2 reports descriptive statistics for the two groups and Figure 4 plots their earnings paths around the event.

¹⁵Given X_i , both the pre-event tax bracket and the instrument of equation (4) are determined. The idea of the cell design is to leave partner income as the only source of variation in Z_i within a cell.

Table 2: Predicted-treated and predicted-control individuals in one cell

	Women		Men	
	Pred. control (N=2,027)	Pred. treated (N=1,755)	Pred. control (N=4,684)	Pred. treated (N=2,576)
Work income	34,909	34,922	34,952	35,027
Partner's work income	23,109	30,372	21,297	28,943
Age	53.2	53.8	54.0	54.7
Partner's age	54.2	54.9	52.1	53.1
Change in marginal tax rate, predicted	0.00	0.16	0.00	0.16
Change in marginal tax rate, actual	0.02	0.11	0.02	0.10

Notes: The cell comprises individuals with pre-event own income in [€34,000, €36,000], a pre-event marginal tax rate of 14%, and 2.5 fiscal parts, whose first-born exits the tax unit at $r = 0$. Predicted controls are individuals for whom the event is not predicted to change the marginal tax rate ($Z_i = 0$); predicted treated are individuals pushed into the 30% bracket. All variables are measured in the pre-event year $r = -1$, except the predicted and actual tax changes, which are measured between $r = -1$ and $r = 1$. The two groups are balanced on own income and age, and differ in partner income, which is what determines whether the fall in the family quotient pushes per-part income across the threshold of the post-event schedule. The tax-change rows are expressed in marginal-tax-rate points; the predicted row corresponds to the bracket transition underlying the instrument of equation (4), whose value in log net-of-tax units is $Z_i = \ln(0.70/0.86) = -0.206$ for the 14%-to-30% transition, and the actual row is the realised change in the household marginal tax rate.

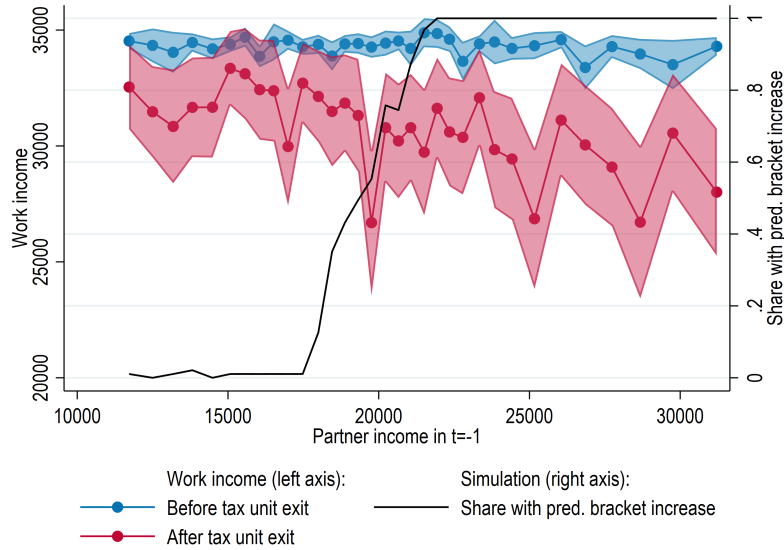


Figure 3: Identifying variation at the first-born's fiscal exit

Notes: Sample: women in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The probability of a predicted tax increase rises sharply with partner income, dividing the sample into a validation region, in which the tax unit experiences the exit but no change in tax bracket, and a treatment region, in which the exit is predicted to raise the marginal tax rate. Identification comes from the difference between the two regions; the validation region provides a direct check that partner income does not by itself predict earnings growth. The figure follows the graphical validation approach of Jakobsen and Sogaard (2022). The corresponding figure for men is Figure B.1 in Appendix B.

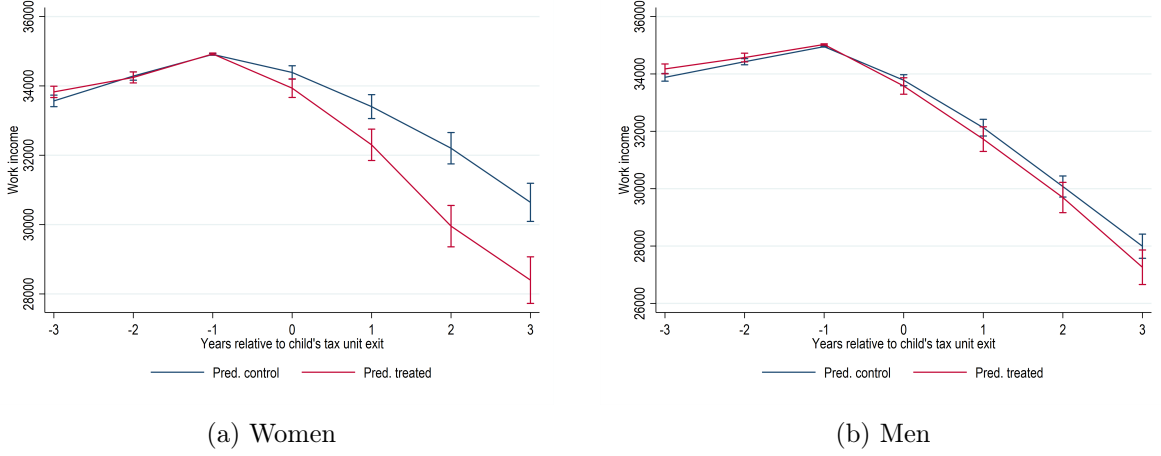


Figure 4: Earnings around the first-born's fiscal exit, by predicted treatment status

Notes: Cell-level raw means for the cell of Table 2, without matching or covariate adjustment, so the figure previews exactly the comparison the design uses. Labour income is measured relative to the pre-event year $r = -1$; the vertical line marks the event. The series distinguish individuals predicted to change tax bracket at the event ($Z_i = \delta < 0$) from those predicted not to ($Z_i = 0$).

The remainder of this section states precisely what such comparisons identify.

4.2 Potential earnings model

To interpret these comparisons, we cast them in a potential earnings model that links them to individual elasticity parameters.

Let $D_{ir} \in \{0, 1\}$ indicate whether individual i 's household is taxed under the post-event fiscal configuration at event time r ; mechanically, $D_{ir} = \mathbf{1}[r \geq 0]$, and in the notation of Section 3, $D_{ir} = g_{i,m_i+r}$; we keep a separate symbol for its distinct role here, as the treatment indicator of the mapping below. Given the two configurations, we let $Y_{ir}(d)$ denote potential log earnings at event time r under configuration d , and $\text{NTR}_{ir}(d)$ the corresponding potential log net-of-tax rate.¹⁶ The potential outcomes map to observed outcomes through¹⁷

$$\begin{aligned}
 (7) \quad Y_{ir} &= Y_{ir}(0) + D_{ir} \mathbf{1}[Z_i \neq 0] [Y_{ir}(1) - Y_{ir}(0)], \\
 \text{NTR}_{ir} &= \text{NTR}_{ir}(0) + D_{ir} \mathbf{1}[Z_i \neq 0] [\text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)].
 \end{aligned}$$

¹⁶ $Y_{ir}(0)$ is not a no-event counterfactual: the event occurs under both configurations, and only the fiscal configuration is varied.

¹⁷For the predicted-control group the mapping sets observed outcomes equal to untreated potential outcomes at all horizons; the control group's realised net-of-tax-rate changes—its first stage—provide a direct check (Appendix B).

Following Saez et al. (2012) and Graber et al. (2026), we assume the potential earnings function is

$$(8) \quad Y_{ir}(\text{NTR}, d) = \epsilon_i \times \text{NTR} + \nu_{ir}(d),$$

where ϵ_i is the individual's elasticity parameter—interpreted through the model of Section 3 as the total household-rate elasticity, $\epsilon_i = \epsilon_i^H = \epsilon_{ii} + \epsilon_{ip}$ —and $\nu_{ir}(d)$ collects all determinants of log earnings other than the marginal net-of-tax rate. It is useful to decompose

$$(9) \quad \nu_{ir}(d) = \mu_{ir} + \xi_{ir}(d), \quad \xi_{ir}(0) \equiv 0.$$

The component μ_{ir} varies with event time and collects everything that evolves over the event window irrespective of the fiscal configuration: life-cycle and calendar-time earnings growth and, in particular, the direct, non-tax consequences of the event itself—changed time use, home production, transfers. Because the event occurs under both configurations (footnote 16), these direct effects cannot differ across them and belong to μ_{ir} . The component $\xi_{ir}(d)$ collects any channel through which the fiscal configuration itself affects earnings other than the marginal rate, such as the change in tax liability.

We impose three assumptions.

Assumption 1 (Conditional common trends). *For every cell, horizon r , and dose δ in the within-cell support of Z ,*

$$\mathbb{E}_c[\Delta_r Y(0) \mid Z_i = \delta] = \mathbb{E}_c[\Delta_r Y(0) \mid Z_i = 0], \quad \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = \delta] = \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = 0].$$

Assumption 1 conditions on the cell and contrasts the dose, in the form familiar from conditional difference-in-differences (Abadie, 2005; Callaway & Sant'Anna, 2021; Heckman et al., 1997). It holds, in particular, whenever untreated growth does not vary within cells at all, $\mathbb{E}[\Delta_r Y(0) \mid X_i = x] = f_y^r(c(x))$ for all x , and likewise for $\Delta_r \text{NTR}(0)$: since own income, the event year, and the event type are fixed by the cell, this requires that the exact position of partner income within the cell's partner-income band does not predict untreated growth. By the potential earnings function and (9), $\Delta_r Y(0) = \epsilon_i \Delta_r \text{NTR}(0) + \Delta_r \mu_{ir}$, so the restriction

concerns the untreated evolution of net-of-tax rates and of the non-tax determinants μ_{ir} —the latter including the direct effects of the event. Because Assumption 1 is stated for all horizons, its pre-event implications are testable (Section 4.5).¹⁸

Assumption 2 (Exclusion restriction). $\nu_{ir}(0) = \nu_{ir}(1)$ with probability one; equivalently, $\xi_{ir}(1) = 0$ in (9).

The configuration change affects earnings only through the marginal net-of-tax rate: it is Assumption S1 of Section 3.1 in potential-outcomes form.¹⁹ Within the model of Section 3, it holds if and only if there are no income effects, in which case ϵ_i is the compensated elasticity, which coincides with the uncompensated one (Graber et al., 2026, Propositions 3–4); with income effects, the estimands recover a total response to the event-induced change in the tax schedule. The event-induced liability change, however, scales with the amount of income shifted across a bracket threshold and can be predicted from pre-event incomes in the same way as the rate change itself; Appendix A uses this second, event-based instrument to identify income effects separately.

Assumption 3 (Monotonicity). $\text{NTR}_{ir}(1) \leq \text{NTR}_{ir}(0)$ with probability one, for all $r \geq 0$.

The condition is stated on the event-induced component of rates—the individual first-stage response $\Phi_{ir} \equiv \text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)$: the configuration change raises no household’s net-of-tax rate. The events we use predict weakly higher marginal rates for every treated individual ($Z_i \leq 0$; Section 3.2), and the condition requires realised first-stage responses not to move against this prediction.²⁰ This is the standard no-defiers condition ensuring that Wald estimands reflect positively weighted averages of individual parameters (Angrist & Imbens, 1995; Imbens & Angrist, 1994). Appendix F shows that the analogous condition is easily violated in reform settings, where income effects and extensive-margin responses can move earnings across brackets against the reform’s prediction while being themselves caused by the reform. Violations are less likely when, as here, the configuration change shifts the thresholds of the schedule rather than its gradient. A violating household must re-cross the shifted threshold

¹⁸Narrower partner-income bands weaken the requirement, and the matching implementation of Section 4.5 additionally aligns pre-event income paths within cells.

¹⁹In terms of (9), Assumption 2 restricts only the configuration-specific component, $\xi_{ir}(1) = 0$; the direct effects of the event enter through μ_{ir} , which carries no configuration argument and remains unrestricted.

²⁰Only variation caused by the event is restricted: realised rates may move in either direction through the untreated component $\Delta_r \text{NTR}(0)$ without violating the condition.

through its earnings response alone: when the quotient falls from 2.5 to 2, per-part income rises mechanically by a quarter at unchanged earnings, so joint taxable income must fall by more than 20 per cent to undo it. Partner income, which is not mechanically changed by the event, places a floor under this adjustment—an individual earning half of joint income would have to cut her own earnings by more than 40 per cent. The appendix formalises the argument with a simulation.

4.3 Identification

For each cell, dose δ , and horizon, the reduced-form and first-stage contrasts are

$$\begin{aligned} RF_c^r(\delta) &\equiv \mathbb{E}_c[\Delta y_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta y_{ir} \mid Z_i = 0], \\ FS_c^r(\delta) &\equiv \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = 0], \end{aligned}$$

and our headline estimand is the cell–dose elasticity

$$\beta_c^r(\delta) \equiv \frac{RF_c^r(\delta)}{FS_c^r(\delta)},$$

the disaggregated counterpart of β^{DD} .²¹

Proposition 1 (Causal interpretation). *Under the mapping (7), the potential earnings function (8), and Assumptions 1–3, for every cell, dose $\delta \neq 0$, and horizon $r \geq 0$ with $FS_c^r(\delta) \neq 0$:*

$$\beta_c^r(\delta) = \mathbb{E}_c[\omega_{ir} \epsilon_i \mid Z_i = \delta], \quad \omega_{ir} \equiv \frac{\Phi_{ir}}{\mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]},$$

with weights $\omega_{ir} \geq 0$ averaging to one: the cell–dose elasticity is a positively weighted average of individual elasticity parameters among the predicted treated.

Proof. By (7), (8) and Assumption 2, $\Delta y_{ir} = \Delta_r Y_i(0) + D_{ir} \mathbb{1}[Z_i \neq 0] \epsilon_i \Phi_{ir}$ and $\Delta \text{NTR}_{ir} = \Delta_r \text{NTR}_i(0) + D_{ir} \mathbb{1}[Z_i \neq 0] \Phi_{ir}$. By Assumption 1, the conditional means of the untreated components are equal for the $Z_i = \delta$ and $Z_i = 0$ groups of the same cell, so they cancel in the contrasts: $RF_c^r(\delta) = \mathbb{E}_c[\epsilon_i \Phi_{ir} \mid Z_i = \delta]$ and $FS_c^r(\delta) = \mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]$. Taking the ratio gives

²¹Dividing the reduced form by the predicted change instead, $RF_c^r(\delta)/\delta$, scales the response by the intended rather than the realised rate change—the policy elasticity of Hendren (2016); we use this reduced-form variant where noted.

the weighted-average representation, with weights averaging to one by construction; under Assumption 3 all Φ_{ir} are non-positive, hence share the sign of their conditional mean, and the weights are non-negative. $\square$ ²²

Remark (extensive margin). The construction above is stated for the intensive margin. For the extensive margin, the outcome is the participation indicator $P_{ir} \equiv \mathbb{1}\{z_{ir} > 0\}$, in changes relative to the pre-event year, $\Delta_r P_i \equiv P_{ir} - P_{i,-1}$; the same cell-dose contrasts and aggregation apply, and the resulting ratio is a semi-elasticity of participation with respect to the household net-of-tax rate.

4.4 Aggregation

We aggregate in two steps. Within a cell, the dose-specific elasticities are combined with frequency weights over the predicted treated,

$$\beta_c^r \equiv \sum_{\delta \in \mathcal{D}_c} w_c(\delta) \beta_c^r(\delta), \quad w_c(\delta) \equiv \Pr_c(Z_i = \delta \mid Z_i \neq 0),$$

where $\mathcal{D}_c$ denotes the within-cell support of non-zero doses. In the second step, we aggregate all cells contained in a group g of interest—own-income intervals for the income profiles, and own-by-partner-income rectangles for the heatmaps of Section 5:

$$\epsilon_g^r \equiv \sum_{c \in g} v_g(c) \beta_c^r,$$

where $v_g(c)$ is cell c 's share of the group's predicted-treated individuals. We report ϵ_g^r as an event-time profile; the headline estimates of Section 5 fix the reference horizon $r^* = 1$.

Because both steps use non-negative frequency weights, the summary parameters inherit the causal interpretation of Proposition 1: substituting, ϵ_g^r is a weighted average of the individual parameters ϵ_i among the predicted treated of the group, with weights proportional to $v_g(c) w_c(\delta) \omega_{ir}$. These weights are known sample frequencies rather than estimator-implied—in contrast to pooled regressions across treatment intensities, whose implicit weights can turn

²²Proposition 1 is the analogue, for event-based instruments, of the characterisation in Graber et al. (2026) of when reform-based ETI estimands are causal: their two conditions—a binary instrument contrast, and income controls rich enough to reproduce the conditional mean of the instrument—appear here as a feature of the design (each estimand contrasts $\mathbb{1}\{Z_i = \delta\}$ against the zero-dose group) and as Assumption 1.

negative under heterogeneity (Callaway et al., 2024; Graber et al., 2026)—and alternative policy-relevant weightings can be substituted without affecting the identification argument. Assumption 1 restricts untreated growth only *within* cells; nothing restricts how elasticities vary *across* cells.

4.5 Estimation and inference

All parameters are estimated separately for women and men. For each cell, we estimate by least squares the regression

$$(10) \quad \Delta y_{ir} = a_c^r + \sum_{\delta \in \mathcal{D}_c} RF_c^r(\delta) \mathbb{1}\{Z_i = \delta\} + u_{ir}, \quad i \in \mathcal{I}_c,$$

and identically for ΔNTR_{ir} , which yields the $FS_c^r(\delta)$; the cell-dose elasticity is the ratio of the two coefficients, numerically identical to two-stage least squares with instrument $\mathbb{1}\{Z_i = \delta\}$ on the subsample restricted to that dose and the zero-dose group, and the coefficients equal the group-mean contrasts of Section 4.3. Estimating (10) at pre-event horizons $r < 0$ produces the placebo contrasts implied by Assumption 1, which we report alongside the post-event estimates.

The matching implementation replaces the regression comparison within each cell with nearest neighbours on the Mahalanobis distance of pre-event work income, measured at $r = -3$, $r = -2$ and $r = -1$, with exact matching on the event year and a caliper of 0.5; RF , FS and β are then matched differences in means and their ratio (footnote 18).

Standard errors are obtained by the delta method applied to the estimated (RF, FS) pairs, with variances clustered at the household level; cells are treated as independent, and the same delta-method step propagates to the aggregates β_c^r and ϵ_g^r , whose weights are sample frequencies with negligible estimation error. Cell-dose comparisons with fewer than $\underline{N}$ treated or $\underline{N}_0$ zero-dose individuals are excluded from estimation and aggregation.

5 Results

This section reports the results. Cell-level elasticities are estimated from equation (10) and aggregated with the weights of Section 4.4: first to population averages, then along the own-income distribution (Section 5.1), and finally over the joint distribution of own and partner income (Section 5.2).

Population averages An advantage of the disaggregated approach is that the cell-level estimates can be aggregated to any target population. Weighting the own–partner income cells by their frequency in the French population and pooling genders yields an average earnings elasticity of X (X for women, X for men).²³ With income weighting,²⁴ the average elasticity is X .

5.1 Heterogeneity by own income

Figure 5 reports elasticities aggregated within own-income groups, at horizon $r = X$. For women (Panel A), elasticities are largest at the bottom of the distribution, at around X , and decline with own income to X . For men (Panel B), the profile is X .

²³Measured X years after the event.

²⁴as is common because of x

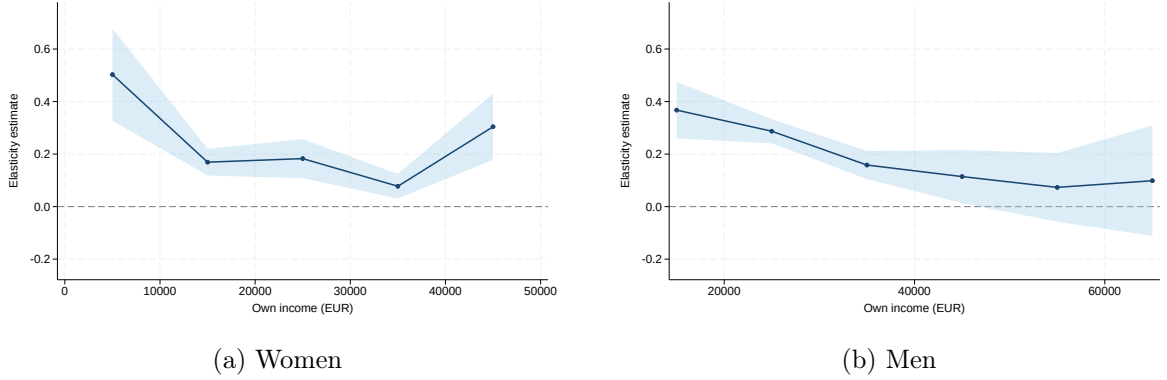


Figure 5: Intensive-margin elasticities along the own-income distribution

Notes: Sample: individuals in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The outcome is earnings growth relative to $r = -1$, pooled over event years $r = 0$ to 3; the pooling window is identical across the heat maps and income profiles, and the event-time profiles of Appendix E display each year separately. Estimates are obtained as described in Section 4: cell-dose elasticities (residualised-IV specification) are aggregated within own-income groups with inverse-variance weights. Cells enter under the same screens as the heatmaps (standard error at most one, first-stage $t \geq 3$, pooled pre-period insignificance); in addition, cells with an elasticity-scale standard error above 0.5 are excluded and income groups supported by fewer than three cells are not displayed. The horizontal axis is own pre-event income, measured at $r = -1$; shaded areas are 95% confidence intervals.

Two pieces of complementary material accompany Figure 5. Appendix E shows how the displayed horizon compares to other horizons and to placebo estimates for the pre-event years. Appendix X reports the number of observations and the standard error underlying each cell; the interior of the income distribution is estimated more precisely than its edges.

The profiles are robust to replacing the regression design with nearest-neighbour matching (Appendix Figure X).

5.2 Heterogeneity by own and partner income

The own-income profile aggregates over households that differ in how income is divided within the couple. Figure 6 disaggregates it, reporting one elasticity estimate per own-income bin and within-bin partner-income tercile. The horizontal axis is own income; the vertical axis orders partner income *within* each own-income bin, so that the comparison across terciles holds the own-income composition fixed.

The map shows that the negative own-income gradient of Figure 5 is present across the partner-income distribution: responses are concentrated at low own income in every partner tercile,

and decline with own income within each tercile. Beyond the event year the concentration shifts towards individuals with higher-earning partners: pooling $r = 1$ to 3, the low-own-income estimate is 0.25 in the top partner tercile against 0.08 in the bottom tercile, and women’s pooled response (0.18) exceeds men’s (0.16).

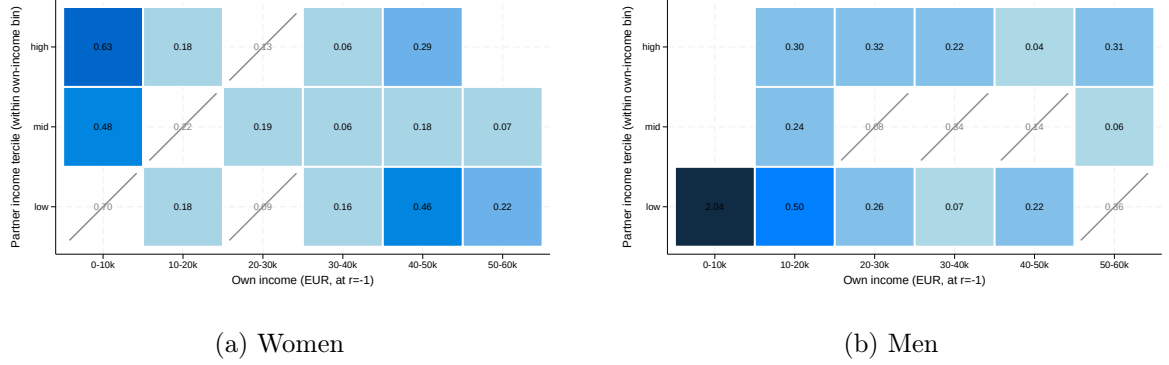


Figure 6: Intensive-margin elasticities by own and partner income

Notes: Sample: individuals in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The outcome is earnings growth relative to $r = -1$, pooled over event years $r = 0$ to 3; the pooling window is identical across the heat maps and income profiles, and the event-time profiles of Appendix E display each year separately. Each tile is one estimate: cell-dose elasticities (residualised-IV specification, matching the income profiles) are pooled with inverse-variance weights within the own-income bin $\times$ partner-tercile group, under the same screens as the income profiles (standard error at most one, first-stage $t \geq 3$, pooled pre-period insignificance, elasticity-scale standard error of the pooled estimate at most 0.5). Printed numbers are the estimates. Tiles crossed out in grey carry estimates that fail their own pre-period placebo (pooled pre-period $|t| \geq 1.96$), or whose placebo cannot be computed; their values are printed in grey for completeness but excluded from the colour scale and from all pooled statistics. Blank tiles are not estimable in the underlying data or fail the precision screens. Both incomes are measured at $r = -1$; the partner terciles are data-driven and their euro boundaries vary across own-income bins.

Figure 7 repeats the display for the extensive margin. Participation responses show the same concentration at low own income, present in every partner tercile.

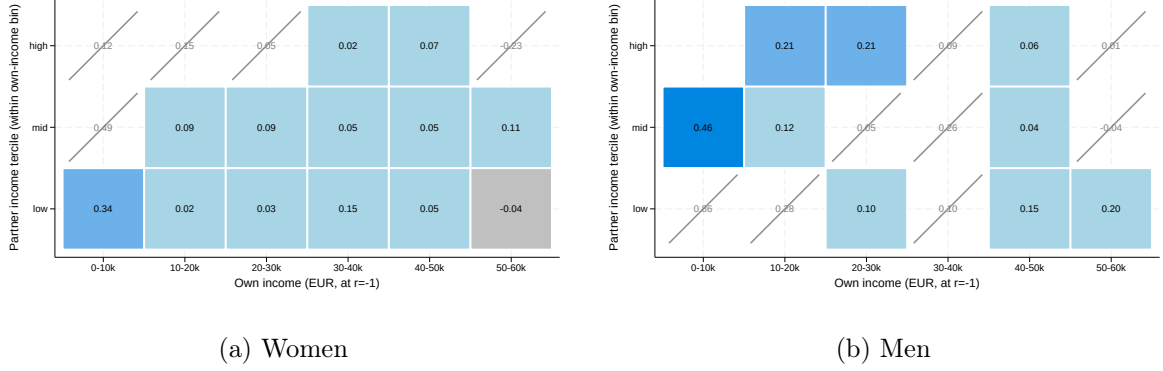


Figure 7: Extensive-margin elasticities by own and partner income

Notes: As Figure 6, for the extensive margin: the outcome is an indicator for positive work income, and tile estimates are participation elasticities constructed as described in Section 4. Sample, pooling window ($r = 0$ to 3), screens, and axes are identical to the intensive-margin display.

6 On the relevance of accounting for heterogeneity

Section 5 documented substantial heterogeneity in estimated elasticities, both across the income distribution and by the composition of income within the household. In this section, we examine whether this heterogeneity matters for tax design. The analyses in this section are, at this stage, illustrative: they are run with placeholder elasticity inputs that mimic the shape of our findings, not yet with our estimates.

We assess this question against two methodologically distinct benchmarks. The first is the normative optimum: the Mirrlees–Saez tradition solves for the tax schedule that maximises a social welfare function subject to resource and incentive constraints (Mirrlees, 1971; Saez, 2001), and its counterpart for couples solves the corresponding multidimensional screening problem (Golosov & Krasikov, 2025; H. J. Kleven et al., 2009). The second is the evaluation of reforms: the perturbation approach of Bierbrauer et al. (2024) takes the observed tax system as given and derives sufficient statistics under which a move towards more individualised taxation of couples is Pareto- or welfare-improving. Section 6.1 conducts the first exercise using our estimates of tax elasticities by own income; Section 6.2 conducts the second using our results on heterogeneity in tax responsiveness by the division of income within couples.

6.1 In optimal tax models

The modern nonlinear-income-tax framework originates with Mirrlees (1971). A government chooses a tax schedule to maximise a social welfare function subject to its resource constraint and to incentive compatibility; because ability is unobserved, the schedule can condition on earnings but not directly on ability. Saez (2001) expresses the resulting optimality conditions in terms of empirically meaningful sufficient statistics: behavioural elasticities, the earnings distribution, and social marginal welfare weights. Importantly for our purposes, these conditions are local. The optimal marginal rate at earnings level z balances the mechanical revenue and welfare effects of a small rate change against the behavioural response of taxpayers at z , which is governed by the elasticity *at* z . Nothing in the formula requires this elasticity to be common across the distribution; the constant-elasticity calibrations used in practice are a convenience rather than a feature of the theory.²⁵

We adapt the sufficient-statistics implementation of Brewer et al. (2010) and replace its common elasticity with a profile $e(z)$. Figure 8 provides an illustrative comparison. The dashed line assumes a constant elasticity of 0.326, the mean of the heterogeneous profile; the solid line instead uses a smooth elasticity profile that declines from 0.4 at low earnings to 0.1 at high earnings. Under both specifications the schedule is U-shaped over the displayed range. The comparison isolates the pointwise inverse-elasticity logic of the formula: the higher elasticity at the bottom lowers optimal marginal rates there, while the lower elasticity higher in the distribution raises them, so the heterogeneous-elasticity schedule is substantially steeper than the constant-elasticity benchmark. A single elasticity calibrated to an economy-wide average accordingly misstates optimal rates at both ends of the distribution.

²⁵The sufficient-statistics formula of Saez (2001) already allows the elasticity to vary with earnings. We evaluate $e(z)$ pointwise while holding the earnings distribution, the social marginal welfare weights, and the revenue requirement fixed. The current figure uses an illustrative smooth profile declining from 0.4 to 0.1; in the final application, this profile will be replaced by an interpolation of our estimated income-group elasticities.

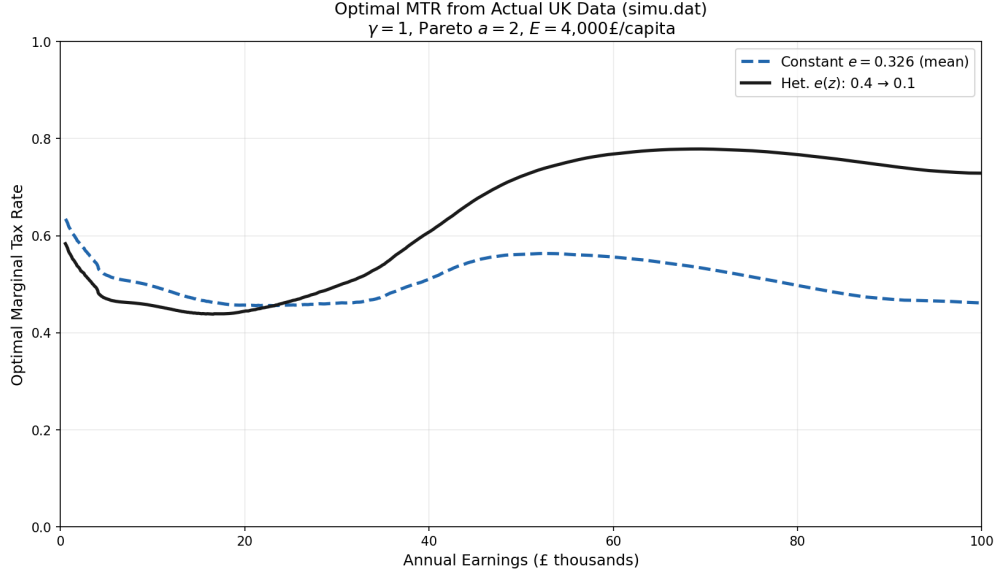


Figure 8: Optimal marginal tax rates under constant and heterogeneous elasticities

Notes: The figure plots optimal marginal tax rates from the sufficient-statistics implementation of Brewer et al. (2010). The dashed line assumes a constant elasticity of 0.326, the mean of the heterogeneous profile; the solid line uses an illustrative smooth profile declining from 0.4 to 0.1. The calibration uses the UK earnings distribution, logarithmic social welfare ($\gamma = 1$), a Pareto-tail parameter of $a = 2$, and a revenue requirement of £4,000 per capita.

This exercise exploits only the variation of responses with own earnings. By construction, a one-dimensional formula has no notion of a partner and therefore cannot absorb the variation of elasticities with the composition of household income. Using that dimension requires a framework for couples, to which we now turn.

6.2 In the assessment of jointness reforms

For couples, solving for the optimal two-dimensional tax schedule is a multidimensional screening problem, which is analytically demanding even under strong simplifying assumptions (Goloso & Krasikov, 2025; H. J. Kleven et al., 2009). In particular, the analytical characterisation in Goloso and Krasikov (2025) relies on a constant elasticity that is common to all individuals and therefore symmetric across spouses; its results would not survive elasticities that vary with the division of earnings within the couple, an extension that is, to our knowledge, work in progress. We therefore follow Bierbrauer et al. (2024), who evaluate reforms of the *existing* system rather than solving for the optimum. Starting from status-

quo couple taxation, they derive sufficient statistics that determine whether reforms towards more individualised taxation—reforms that allow the marginal rates of the two spouses to differ, lowering them on secondary relative to primary earnings in a revenue-neutral way—are Pareto-improving and, where they are not, whether they raise welfare under given welfare weights.

Their baseline calibration sets the elasticity of primary earnings to 0.25 and that of secondary earnings to 0.75. Our cell-specific estimates $\epsilon_{p,s}$ replace these two constants with an estimated surface: responses vary with the couple’s position in the joint distribution and with the division of earnings within it.

Figure 9 displays the welfare gains from moving towards a more individualised tax system under the constant-elasticity calibration (left column) and under our cell-specific estimates (right column). The first row assigns equal welfare weights to all households; the second uses near-Rawlsian weights that place greater weight on low-income households.

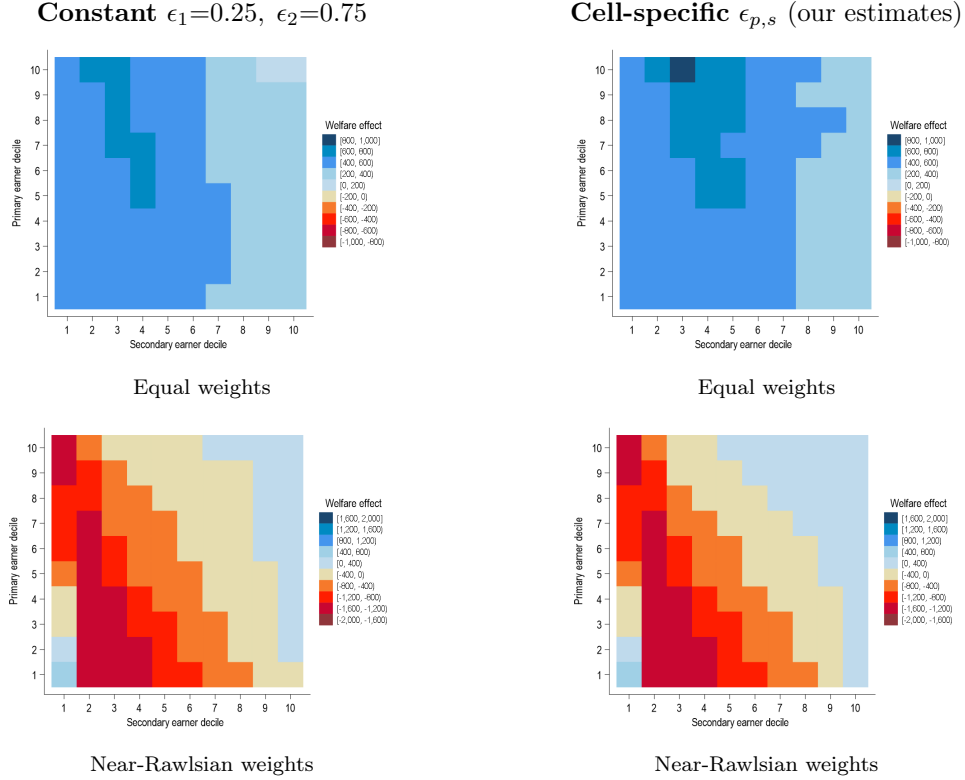


Figure 9: Welfare gains from a more individualised tax system

Notes: The left column uses constant elasticities of $\epsilon_1 = 0.25$ for primary earners and $\epsilon_2 = 0.75$ for secondary earners. The right column uses cell-specific elasticity estimates $\epsilon_{p,s}$. The top row assumes equal welfare weights, while the bottom row assumes near-Rawlsian welfare weights that place greater weight on low-income households. Comparing columns within each row shows that the assessment is sensitive to the interaction between the location of large elasticities in the joint distribution and the welfare weights: the cell-specific estimates raise the assessed gains from individualisation, most markedly under near-Rawlsian weights.

The two exercises show that accounting for heterogeneity changes the shape of the optimal tax schedule and the assessment of reforms towards more individual taxation of couples.^{26,27,28}

²⁶The direction of the optimal primary–secondary differential is itself unsettled. With a common elasticity for all individuals, optimal marginal rates on secondary earners are on average *higher* than on primary earners, because low secondary earnings identify couples to whom a redistributive planner wants to target transfers (Goloso & Krasikov, 2025); the classic argument of Boskin and Sheshinski (1983) instead prescribes *lower* rates on the more responsive spouse. Which force dominates depends on the size and location of the elasticity gap between spouses—the object we estimate.

²⁷Constant-elasticity calibrations are not an innocuous simplification in one dimension either: the welfare assessment of the post-war US tax cuts reverses sign as the assumed constant elasticity of taxable income is varied over plausible values (Bierbrauer et al., 2021); estimated elasticity profiles remove this ambiguity.

²⁸A further margin, which we do not pursue, is political feasibility: with heterogeneous elasticities, assessments of the political feasibility of moves towards individualisation may change (Bierbrauer et al., 2025).

7 Conclusion

This paper proposes a source of identifying variation for estimating tax elasticities, suited to mapping heterogeneities across gender, income and spousal income. Exploiting the institutional features of French family taxation, we use the fiscal exit of a couple’s first-born child—and, as a robustness design, marriage—as an event that changes the mapping from incomes to marginal tax rates, generating variation in marginal incentives through not-yet-tax-liable partner income, under transparently testable assumptions.

Using the universe of French tax declarations from 2003 to 2023 and a panel structure that follows individuals across household-unit changes, we show that headline averages— X for men and X for women—mask substantial heterogeneity. The averages are carried by low own incomes and by women, whose responses exceed men’s, rather than by the remainder of the distribution. Moreover, the negative own-income gradient is driven by individuals insured by a higher-earning partner, not by those in households where both partners earn little.

Feeding the estimates into standard frameworks for tax design changes the conclusions those frameworks reach. The optimal tax schedule implied by the sufficient-statistics formula is substantially steeper than under a constant-elasticity benchmark, and the assessed welfare gains from reforms towards more individual taxation of couples are larger once the constant elasticities such evaluations conventionally assume are replaced with our estimates (Section 6).

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A Model extensions

Income effects

The quasi-linearity of household consumption (Assumption S1) shuts down income effects. Replacing linear consumption with a concave utility $v(c_t)$, the household problem becomes

$$\max_{z_{it}, z_{pt}} v(c_t) - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}; Q_{it}) - \kappa_{it}(g_{it}),$$

where $\kappa_{it}(g_{it})$ collects lump-sum household expenses, which may vary with the event (expenses for the child). The first-order conditions become

$$\frac{\partial \Psi}{\partial z_{jt}} = v'(c_t) \tau_{jt}^H, \quad j \in \{i, p\} :$$

earnings choices now depend on the marginal utility of household consumption, and therefore on partner income, on the level of the tax *liability*, and on needs κ_{it} . Log-linearising adds a virtual-income term to equation (2), in the spirit of Gruber and Saez (2002). Two remarks follow for our design. First, households predicted to cross a bracket at the event also experience a larger increase in tax liability than their within-cell controls, so, absent a virtual-income instrument, our estimates should be read as total (uncompensated) responses to the event-induced tax change; an extension instrumenting a simulated virtual-income term would separate substitution from income responses. Second, the component of the needs change $\kappa_{it}(1) - \kappa_{it}(0)$ that is common within a comparison cell affects treated and control households alike and differences out; only a component covarying with threshold placement would threaten identification, which is the same requirement as for the direct effect on potential income.²⁹

The participation margin

The intensive-margin model cannot generate extensive-margin responses. Following H. J. Kleven et al. (2009), consider a spouse deciding whether to work at all, facing a fixed cost of participation q_{it} (commuting, childcare, forgone home production). Conditional on working,

²⁹Appendix F takes income effects seriously for a different purpose: they are one reason why first-stage monotonicity can fail in reform-based simulated-IV designs, though not in ours.

earnings are chosen optimally, so the value of participation is

$$V_{it}^w = \max_z \left\{ z - [T_t^H(z + z_{pt}; Q_{it}) - T_t^H(z_{pt}; Q_{it})] - \psi(z; n_{it}) \right\},$$

where the bracketed term is the *participation tax*: the increase in the household's tax liability caused by the spouse's earnings, given partner income and the fiscal configuration. The spouse works if and only if

$$V_{it}^w \geq q_{it}.$$

The relevant price on this margin is therefore not the marginal rate at current earnings but the participation tax wedge—and its first-euro marginal rate, $1 - \tau^H$ evaluated at $z = 0$, is exactly what partner income and the quotient determine: because partner income alone places the household above the tax-free threshold, the second earner's first euro is taxed at the household's current bracket. A household pushed into a higher bracket by the child's fiscal exit thus sees the return to the second earner's participation fall at every level of own earnings, which rationalises the extensive-margin responses we document. With heterogeneous fixed costs q_{it} , the extensive-margin analogue of the elasticity is the response of the participation probability to the participation net-of-tax rate.

B Complementary material - case study

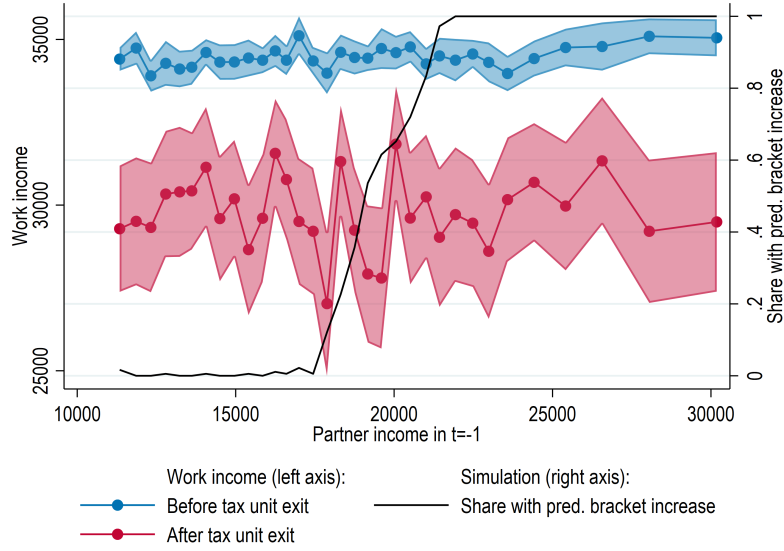


Figure B.1: Identifying variation at the first-born's fiscal exit: men

Notes: Male counterpart of Figure 3. Sample: men in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The probability of a predicted tax increase rises sharply with partner income, dividing the sample into a validation region and a treatment region; see the notes to Figure 3 for construction details.

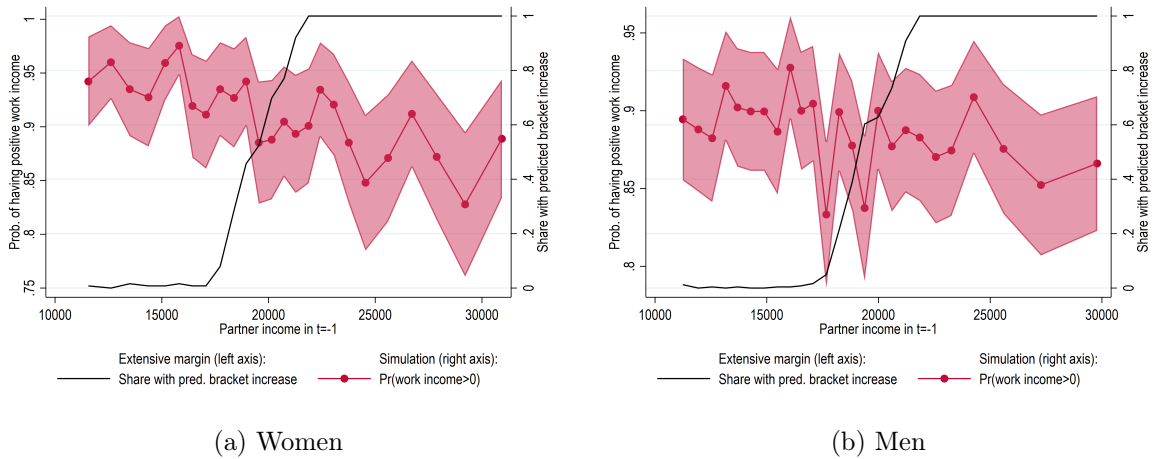


Figure B.2: Identifying variation at the first-born's fiscal exit: extensive margin

Notes: Extensive-margin counterpart of Figures 3 and B.1: the outcome is an indicator for positive work income. Sample and construction as in Figure 3.

Translating these earnings paths into elasticities with the estimation approach of Section 4 yields the second-stage estimates shown in Figure B.3.

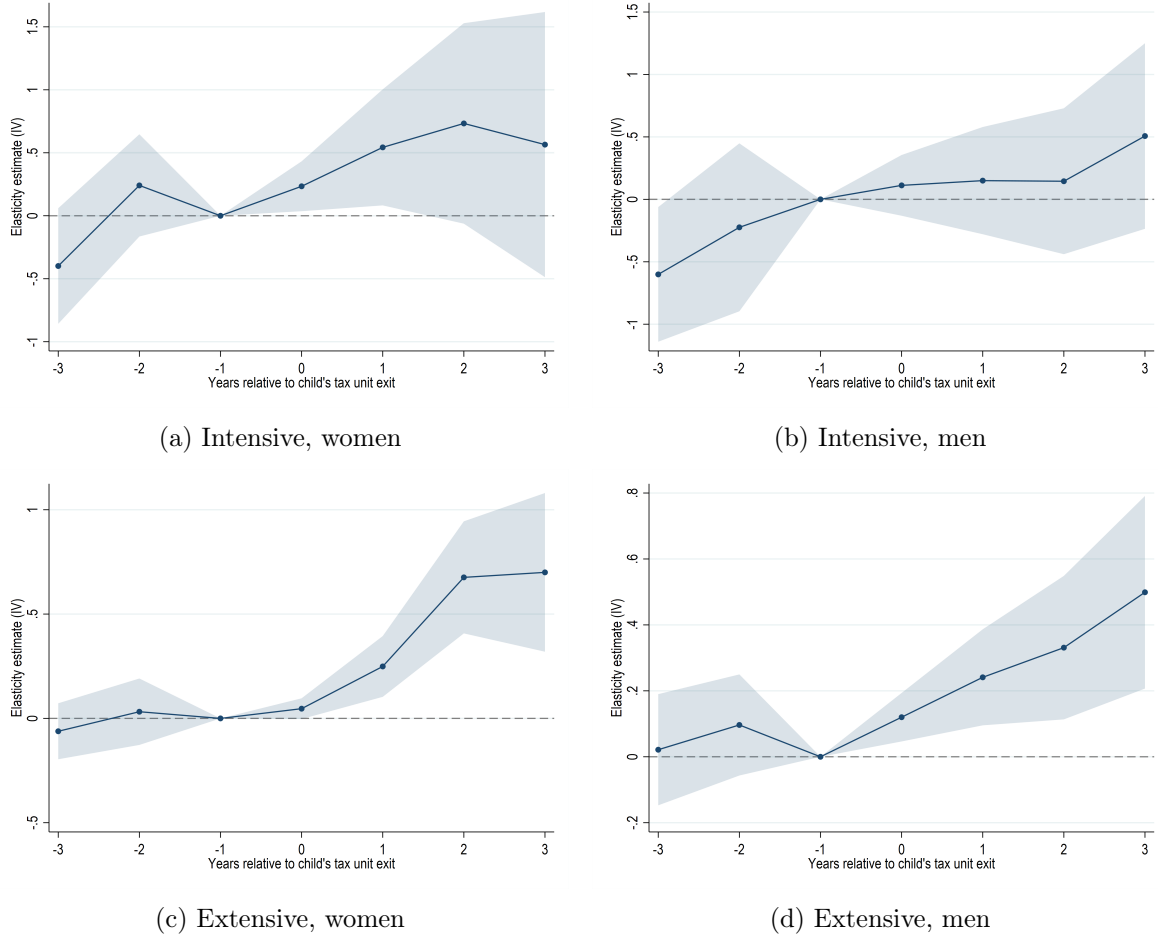
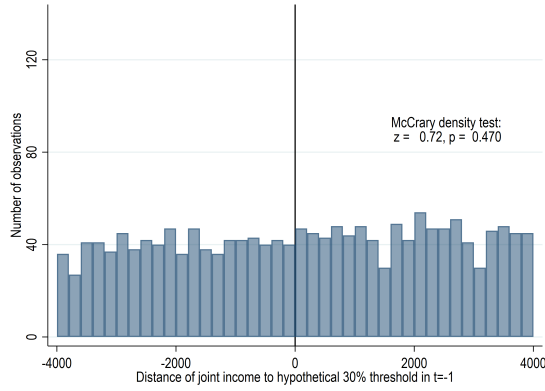
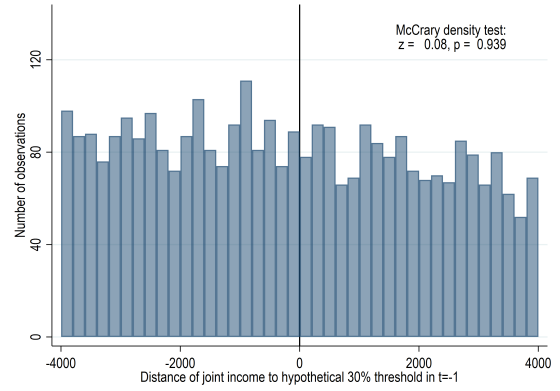


Figure B.3: Second-stage estimates for the case-study cell

Notes: Second-stage (instrumented) estimates for the case-study cell of Table 2, by margin and gender. The instrument, screens, and construction follow Section 4.



(a) Women



(b) Men

Figure B.4: McCrary density test at the hypothetical joint-income threshold

Notes: Each panel plots the number of observations by the distance of pre-event joint income to the hypothetical threshold at which the marginal tax rate rises from 14% to 30% once the first-born's exit reduces the family quotient, for the case-study sample of Table 2, separately for women (Panel A) and men (Panel B). This distance is the running variable underlying the binary instrument: households above the threshold are predicted treated, households below it predicted control. If households anticipated the exit and adjusted incomes strategically—bunching below the future threshold, or timing the detachment—the pre-event density would show excess mass around zero. The distribution is flat through the threshold and the McCrary test does not reject continuity ($z = 0.72$, $p = 0.47$ for women; $z = 0.08$, $p = 0.94$ for men).

C Robustness: the marriage design

D Extensive-margin estimates

E Event-time profiles

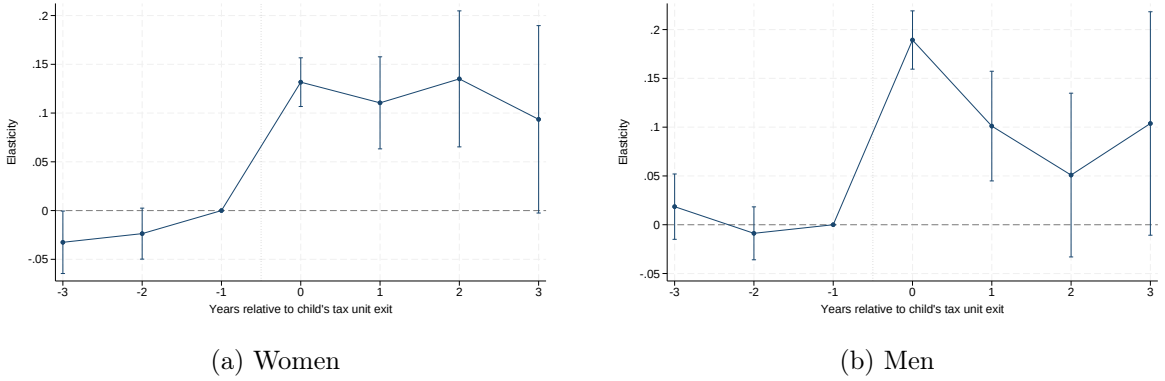


Figure E.1: Intensive-margin elasticities by event time

Notes: Aggregated intensive-margin elasticities by year relative to the first-born's exit from the tax unit, for women (Panel A) and men (Panel B). Each point pools all cell-dose elasticities of the given event year with inverse-variance weights on the elasticity scale; whiskers are 95% confidence intervals. The year before the event, $r = -1$, is the reference and is shown at zero. Estimates for $r = -3$ and $r = -2$ are placebo checks: the event has not yet occurred. Cells enter with the same screens as the heatmaps (standard error at most one, first-stage $t \geq 3$, pooled pre-period insignificance).

F Monotonicity

This appendix (i) states the monotonicity condition in the canonical reform-based simulated-IV setting (Gruber & Saez, 2002; Jakobsen & Sogaard, 2022), (ii) characterises when the condition fails and illustrates a violation in a calibrated example, and, (iii), closes by arguing that violations are far less likely under schedule *shifts*, as in our design, than under reforms that change the schedule's gradient.

The condition in the reform-based setting

Consider a reform that raises statutory rates on part of the schedule. Let $g \in \{0, 1\}$ indicate whether the reform is in force, and let $\text{MTR}_i(g)$ denote the marginal tax rate individual i

faces under regime g , inclusive of behavioural responses. The simulated instrument is the mechanical prediction of $\text{MTR}_i(1) - \text{MTR}_i(0)$, holding income fixed at its pre-reform value. The IV estimand averages individual responses with weights proportional to the realised first-stage responses $\pi_i \equiv \text{MTR}_i(1) - \text{MTR}_i(0)$ (Borusyak et al., 2022; Imbens & Angrist, 1994). For the weights to be non-negative—and the estimand a convex average that cannot reverse the sign of a uniformly signed effect (Blandhol et al., 2022)—the π_i must share a common sign: first-stage monotonicity requires

$$(F.1) \quad \text{MTR}_i(1) \geq \text{MTR}_i(0) \quad \forall i,$$

with the reverse inequality for a reform that cuts rates. A violation—individuals whose realised marginal tax rate moves *against* the direction of the reform—puts negative weight on their responses, and the estimand loses its interpretation as a convex average.

When is it violated, and what does it mean

Under the strong assumptions standard in the literature of (i) a quasi-linear utility model, implying no income effects, (ii) continuous labour supply, and (iii) no heterogeneity in elasticities, condition (F.1) is not violated. The intuition is that, for any reform—whether it increases or decreases the marginal tax rate in a given bracket—the utility ranking between the kink point and the interior solution remains unchanged. As a result, any new interior optimum in a bracket unaffected by the reform is dominated by the kink point at the reform-affected bracket.

When either assumption is relaxed—allowing income effects or labour-supply rigidities—and elasticities or preferences are heterogeneous, first-stage monotonicity can be violated.³⁰

We illustrate a violation of first-stage monotonicity using a model with income effects and heterogeneity in the curvature of the utility function. Instead of the standard quasi-linear utility function, we consider a more general specification that allows for income effects and

³⁰Responses on the extensive margin are a potentially important source of violations. Extensive-margin responses to tax increases are common, as we also document in this paper, and they mechanically reduce the marginal tax rate, as individuals drop out of the labour market and thus move into a lower tax bracket.

non-linear labour supply responses.³¹ Individuals derive utility from consumption c and hours worked h according to

$$u_i(c, h) = \frac{c^{1-\sigma_i}}{1-\sigma_i} - \chi_i \frac{h^{1+\varphi_i}}{1+\varphi_i},$$

where $\sigma_i > 0$ governs the curvature of utility from consumption, and hence the strength of income effects, $\varphi_i > 0$ governs the curvature of the disutility from labour, and $\chi_i > 0$ scales the marginal disutility of hours worked. Labour income is given by $y = wh$. We assume that hours rigidities restrict individuals to four discrete choices: 0, 20, 30, or 40 hours per week.

For the tax schedule, we consider the French income tax system as of 2015. In the part of the schedule on which we focus, labour income up to €26,791 per year is taxed at a marginal rate of 14 %, and income above this threshold at 30 %.³² Because utility is defined over hours worked, we can equivalently express the tax schedule as a function of hours, with a kink at $h = k$ corresponding to the income threshold. In the example, we set the hourly wage to €20.6, so that the threshold at €26,791 corresponds to approximately 25 hours of work per week.

We study a hypothetical reform ($g = 1$) that eliminates the 30 % bracket by raising the marginal tax rate above the threshold from 30 % to 41 %, equal to the next statutory rate in the French schedule.

We consider two preference types, Type A and Type B, and calibrate the model such that both share the same discrete optimum of 30 hours per week under no reform. The key parameter governing the relative strength of income and substitution effects is the consumption-curvature parameter σ_i . Type A is calibrated with $\sigma_A < 1$, so that the substitution effect dominates, whereas Type B is calibrated with $\sigma_B \gg 1$, so that the income effect dominates. For each type, the disutility scale χ_i is chosen to match the discrete optimum of 30 hours per week under no reform. The resulting preference parameters are

$$(\sigma_A, \varphi_A, \chi_A) = (0.696, 0.300, 0.272), \quad (\sigma_B, \varphi_B, \chi_B) = (7.000, 0.300, 9.17 \times 10^{-30}).$$

³¹Hours rigidities would be sufficient to illustrate a violation. To make the argument applicable more broadly, however, we consider a more general model with income effects.

³²To simplify the exposition, we ignore the *décote*, which does not affect the example; results including the *décote* are available on request.

Under no reform, both types choose 30 hours per week, placing them in the 30 % bracket:

$$\text{MTR}_A(0) = \text{MTR}_B(0) = 0.30.$$

The simulated instrument, computed by holding both types' incomes fixed at their 30-hour values, accordingly predicts the same treatment change for both: a rise in the marginal tax rate from 30 % to 41 %, that is, a predicted first-stage response of +0.11.

The optimal hours choices under the reform and under no reform are shown in Figure F.1.³³ Under the reform, the two types' optimal choices diverge:

$$h_A^* = 20 \text{ hours/week}, \quad h_B^* = 40 \text{ hours/week}.$$

Type A chooses 20 hours under the reform, moving below the bracket threshold and facing the lower marginal tax rate of 14 %. Type B chooses 40 hours, moving above the threshold and facing the reform rate of 41 %.

The realised treatments therefore respond to the same shock in opposite directions:

$$\pi_A = \text{MTR}_A(1) - \text{MTR}_A(0) = 0.14 - 0.30 = -0.16 < 0, \quad \pi_B = \text{MTR}_B(1) - \text{MTR}_B(0) = 0.41 - 0.30 = +0.11$$

Type A's marginal tax rate moves against the direction of the reform: despite being exposed to a shock that raises statutory rates—and despite the instrument predicting a tax increase for it—its realised marginal tax rate *falls*, violating condition (F.1). Type B's realised treatment moves with the reform.

First-stage monotonicity is therefore violated. The IV estimand places negative weight on the responses of individuals like Type A, is no longer a convex average of causal effects, and is not weakly causal in the sense of Blandhol et al. (2022): with heterogeneous elasticities it can even reverse the sign of a uniformly signed behavioural response.

³³The optimal hours choice under each regime is determined by maximising $u_i(c, h)$ subject to the piecewise-linear budget constraint. There is no closed-form solution in general, so the optimum is obtained numerically by comparing utility across all candidate solutions: interior first-order-condition roots on each linear segment, the kink, and the corner at zero hours.

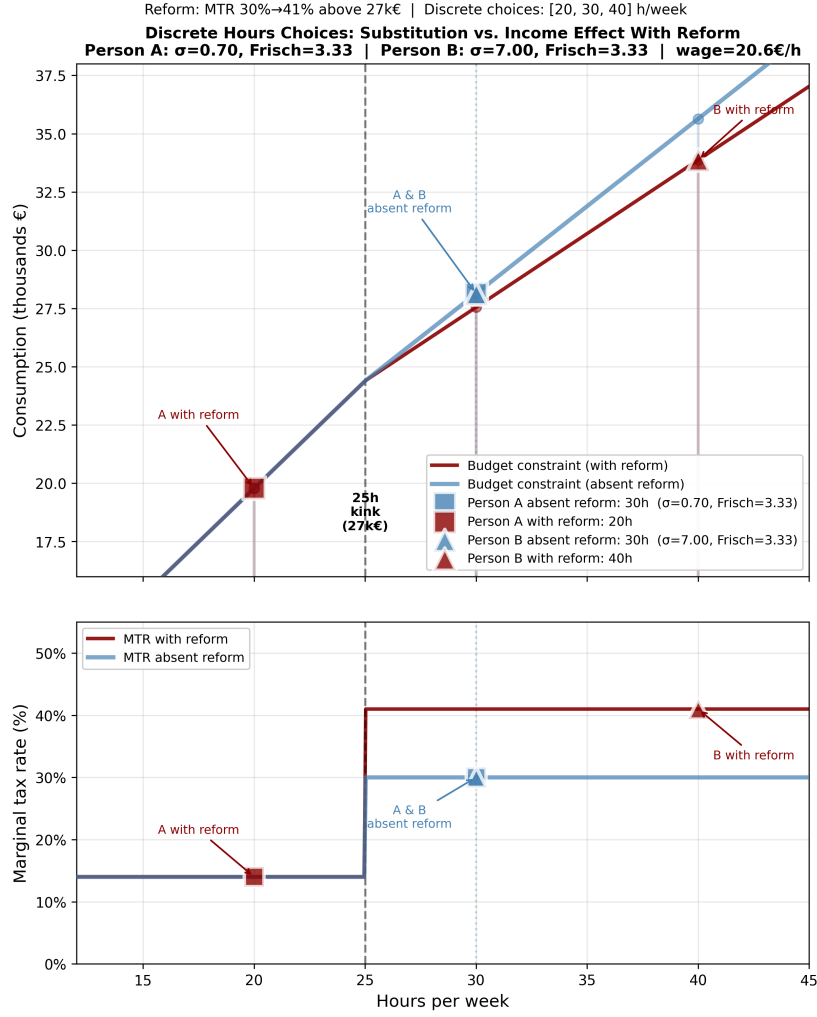


Figure F.1: Violation of first-stage monotonicity under heterogeneous labour-supply responses

Top panel. Budget constraints under the reform (red) and under no reform (blue). The reform raises the marginal tax rate above the second bracket threshold from 30 % to 41 %. Both Type A and Type B optimally choose 30 h/week under no reform. Under the reform, Type A chooses 20 h/week and is below the kink; Type B ($\sigma_B = 7$) chooses 40 h/week and is above the kink.

Bottom panel. Marginal tax rates under the reform (red) and under no reform (blue), as a function of hours worked, with the same optimal choices indicated.

It is worth noting that a strong income effect for Type B is not necessary for the violation. Even in the absence of income effects, Type B could simply choose 30 hours per week under the reform as well and be in the now higher 41 % bracket. The same violation would then arise: Type A chooses 20 hours under the reform and is in the 14 % bracket ($\pi_A = -0.16$), while Type B stays at 30 hours ($\pi_B = +0.11$).

Models with extensive-margin labour supply responses can also generate violations of first-stage monotonicity. An extensive-margin response to, for example, a reform that increases tax rates moves an individual out of the tax schedule altogether, that is, in France, to a marginal tax rate of zero, so that $\pi_i < 0$. If, at the same time, other individuals remain employed and face the higher rate under the reform, $\pi_i > 0$, the realised treatment responds to the shock in opposite directions across individuals, violating condition (F.1). Because extensive-margin responses to tax increases are empirically common, as we also document in this paper, this is a realistic and quantitatively important source of monotonicity violations in reform-based IV strategies.

We stress that the reform-based setting of this example is a foil: it is the canonical design from which our identification strategy departs, not our identifying variation. We now turn to our design.

Why first-stage monotonicity is likely satisfied for our instrument

The argument below is stated for the marriage design; it carries over unchanged to the fiscal exit of a child, which changes the household tax base in the same way.

A key feature of our instrumental variation is that it does not arise from a change in the gradient of the tax schedule—as in the reform-based setting above—but from an outward shift of an otherwise unchanged schedule. Marriage to a higher-earning partner under joint taxation shifts the tax liability function outward, because partner income is added to the household tax base. This means that partner income places a floor on the marginal tax rate an individual faces after marriage, ruling out the bracket-crossing channel through which monotonicity violations arise in the reform-based setting.

Figure F.2 illustrates, for a given level of own income, why such a monotonicity violation is highly unlikely with our instrument. The right panel shows the marginal tax rate an individual faces as a function of own income, both when single (blue dashed) and when married to a partner earning €40,000 (red solid). Crucially, the marginal tax rate for a married individual remains strictly positive—at 14%—even when own income is zero, because the partner’s income already places the household above the tax-free threshold. An extensive-margin response, the most extreme possible reduction in labour supply, therefore still leaves the

individual facing a 14 % marginal tax rate rather than a 0 % rate. Since the instrument in our case study captures individuals who are predicted to move from the 14 % bracket to the 30 % bracket, a violation of monotonicity would require the realised marginal tax rate to fall strictly below 14 % with marriage, which is impossible unless partner income collapses dramatically. The floor imposed by partner income on the married budget set therefore essentially rules out the channel through which monotonicity could be violated, and the logic extends to all individuals facing predicted tax increases. For predicted tax decreases, monotonicity violations are likewise ruled out, since they would require a dramatic increase in labour supply.

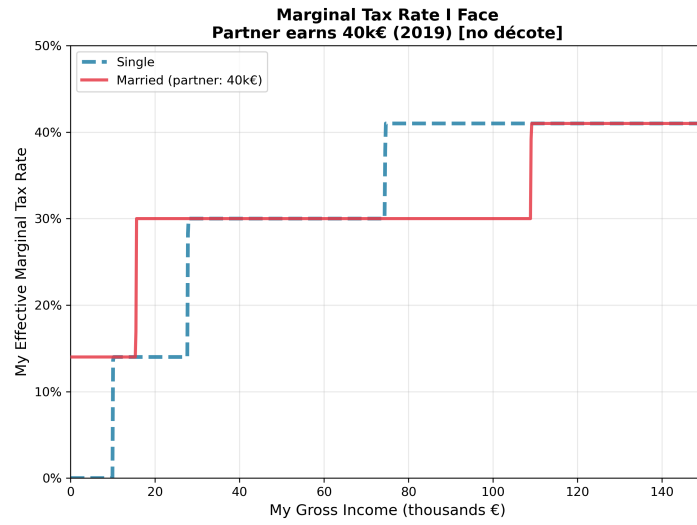


Figure F.2: Individual budget constraint and marginal tax rate under marriage, partner earns €40,000

Notes: The figure shows the effective marginal tax rate as a function of own gross income under individual taxation (blue dashed) and under joint taxation when married to a partner earning €40,000 (red solid), based on the 2019 French income tax schedule without the *décote*. Income is not pooled, so own consumption corresponds to own after-tax income only. As a result, the marginal tax rate for a married individual is 14 % even at zero own income, because the partner's income of €40,000 already places the household above the tax-free threshold. Consequently, even a complete exit from the labour market cannot lower the individual's marginal tax rate below 14 %, ruling out a monotonicity violation for individuals predicted to move from the 14 % to the 30 % bracket.